

PHY202: Problem Set 3

NOTE: Submit the solution to problem number 4 in the beginning of the class on Oct. 6th.

1. Use complex variables to show that the following integral takes value 2π for $n = m$ and vanishes otherwise:

$$\int_0^{2\pi} e^{i(n-m)\theta} d\theta.$$

2. Write down the Fourier series for the periodic function,

$$f(x) = \begin{cases} 0, & -\pi < x \leq 0 \\ \sin x, & 0 < x \leq \pi \end{cases} \quad \text{and } f(x + 2\pi n) = f(x).$$

3. A periodic triangular wave is specified by,

$$f(x) = \begin{cases} -1 - x, & -2 \leq x \leq 0 \\ -1 + x, & 0 \leq x \leq 2 \end{cases} \quad \text{and } f(x + 4n) = f(x).$$

- (i) Expand $f(x)$ as a Fourier series.
- (ii) Find the series for the derivative of the function.
- (iii) Find the series for the integral of the function.

4. Using $f(x) = x^2$ in the interval $-\pi < x < \pi$, show that,

$$(i) \quad \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12},$$

$$(ii) \quad \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6},$$

$$(iii) \quad \sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}.$$